

$$J = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) \quad \left| \quad \vec{L} \equiv \vec{r} \times \vec{p} = \vec{c}$$

$$(r, \theta, \phi)$$

$$\dot{\vec{r}} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + \dot{\phi} r \sin \theta \hat{\phi}$$

$$m \ddot{r} - m r (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = 0$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) - m r^2 \sin \theta \cos \theta \dot{\phi}^2 = 0$$

$$\frac{d}{dt} (m r^2 \sin^2 \theta \dot{\phi}) = 0$$

$$\vec{L} = r \hat{r} \times m (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + r \sin \theta \dot{\phi} \hat{\phi})$$

$$\vdots$$

$$= m r^2 [\dot{\theta} \hat{r} \times \hat{\theta} + \sin \theta \dot{\phi} \hat{r} \times \hat{\phi}]$$

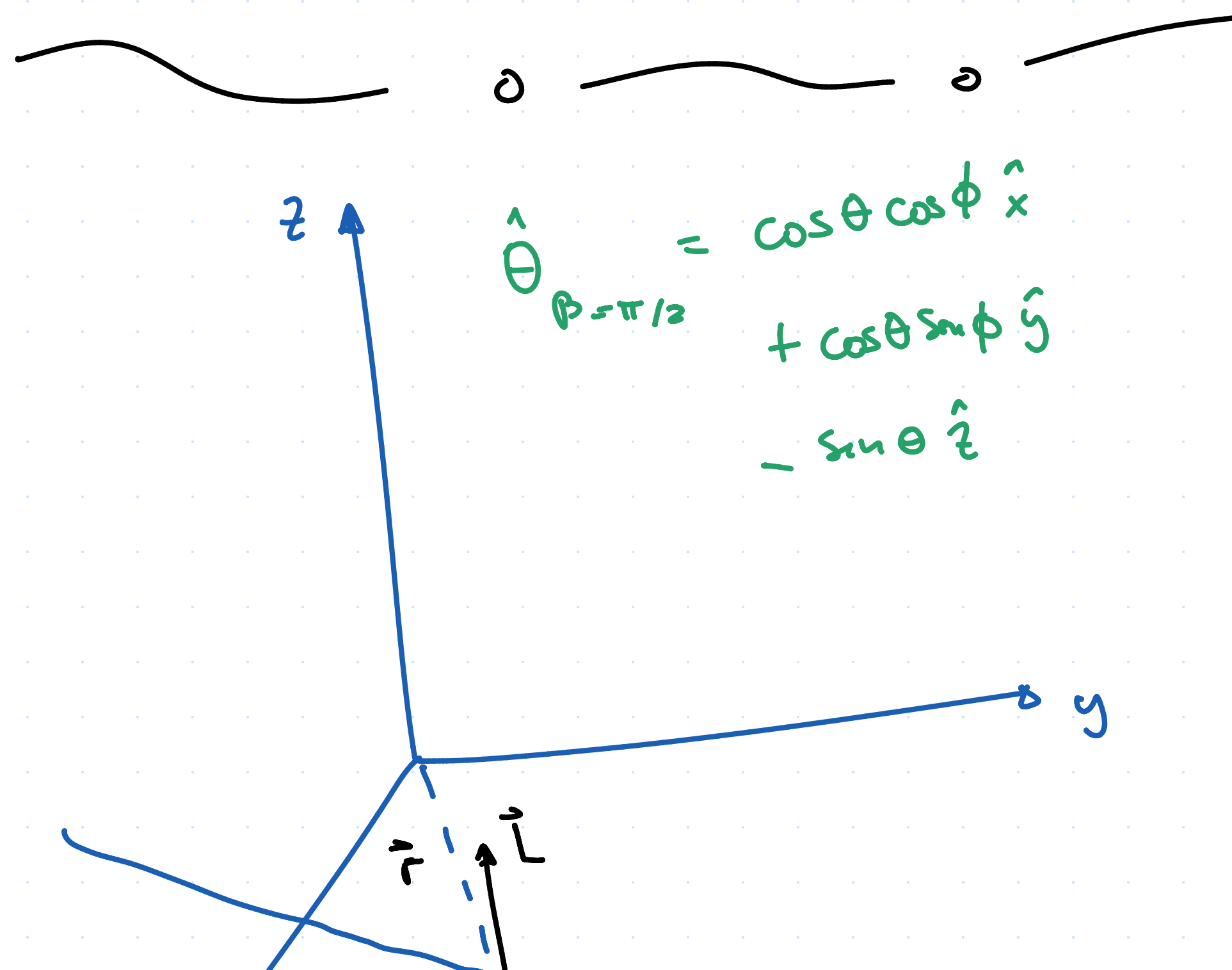
$$\vec{L} = m r^2 [\dot{\theta} \hat{\phi} - \dot{\phi} \sin \theta \hat{\theta}] = \vec{c} \text{ te}$$

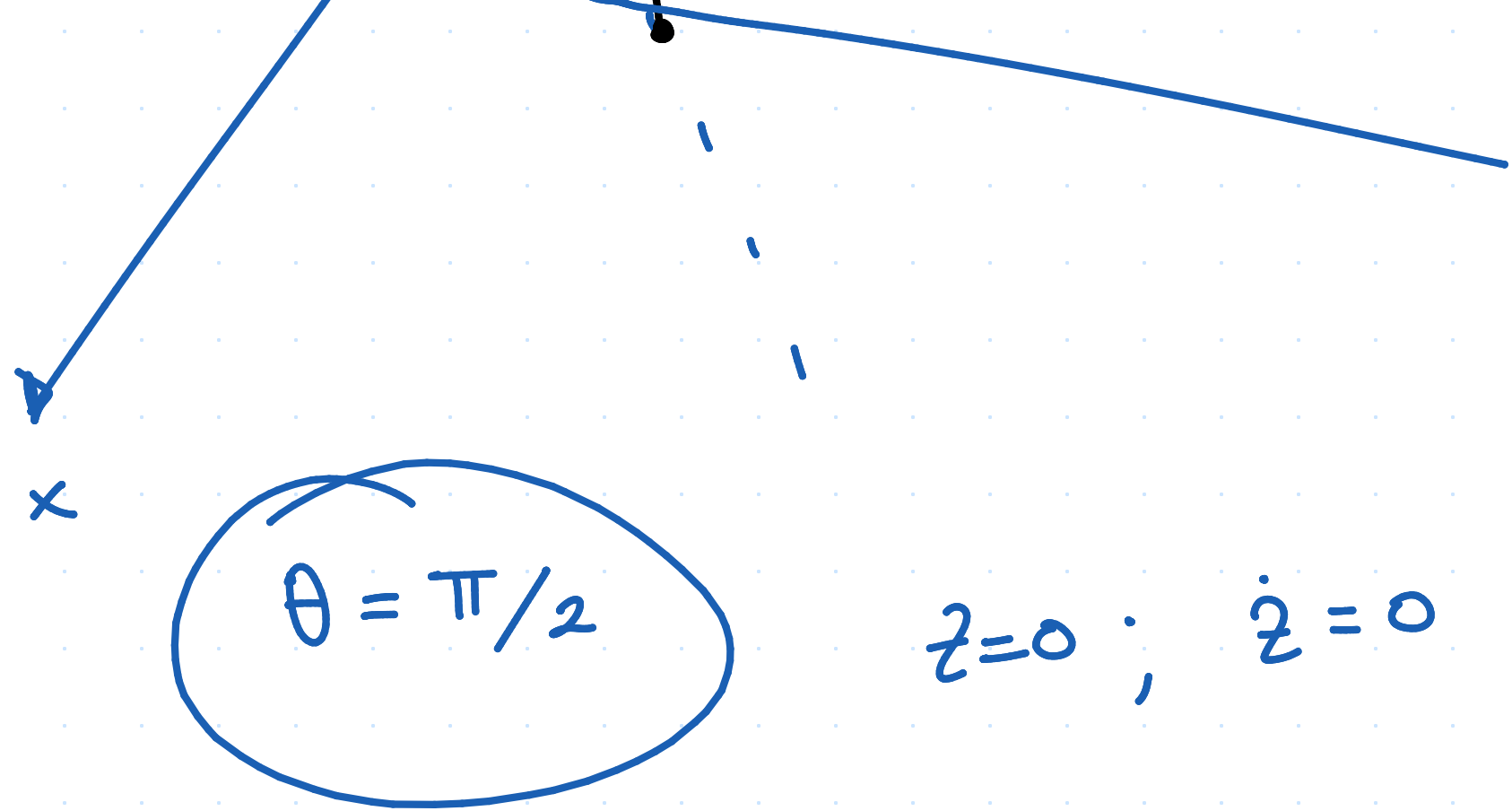
$$L = m r^2 \sqrt{\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta}$$

$$\vec{L} \cdot \hat{z} = L_z$$

$$\vec{L} = -m r^2 \dot{\phi} \hat{\theta} \Big|_{\theta=\pi/2} = m r^2 \dot{\phi} \hat{z} = L \hat{z}$$

$(\theta=\pi/2, \dot{\theta}=0)$





$$z = r \cos \theta \Rightarrow z = 0 \Rightarrow \cos \theta = 0 \Big|_{\pi/2}$$

$$\dot{z} \Big|_{\theta=\pi/2} = -r \dot{\theta} \Big|_{\theta=\pi/2} = 0 ; \dot{\theta} = 0$$

T r e m o s

$$J = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

$$\vec{r}(t) = (x, y, z) = (r, \theta, \phi)$$

$$\boxed{\dot{\theta} = 0 ; \theta = \pi/2} \Rightarrow \theta(t) = \pi/2$$

$$J_0 = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\phi}^2)$$

$$\begin{aligned} r &= r(t) \\ \phi &= \phi(t) \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial J_0}{\partial \dot{r}} \right) - \frac{\partial J_0}{\partial r} = 0$$

$$\frac{d}{dt} \left(\frac{\partial J_0}{\partial \dot{\phi}} \right) - \frac{\partial J_0}{\partial \phi} = 0$$

$$\Rightarrow \frac{d}{dt}(mr\dot{\phi}) - mr\dot{\phi}^2 = 0$$

$$m\ddot{r} - mr\dot{\phi}^2 = 0$$

$$; \quad \frac{d}{dt} \underbrace{(mr^2\dot{\phi})}_l = 0$$

$$mr^2\dot{\phi} = l$$

$$m\ddot{r} - mr\dot{\phi}^2 = 0$$

$$mr^2\dot{\phi} = l$$

$$; \quad E = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2)$$

$$\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2) = E$$

$$mr^2\dot{\phi} = l$$

$$\Leftrightarrow \dot{\phi} = \frac{l}{mr^2}$$

$$\Rightarrow \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \left(\frac{l}{mr^2} \right)^2 = E$$

$$\frac{1}{2} m \dot{r}^2 + \frac{mr^2 l^2}{2m^2 r^4} = E ;$$

$$\left[\frac{1}{2} m \dot{r}^2 + \frac{l^2}{2mr^2} = E \right]$$

Ya no es
 una partícula libre
 (Aparece una interacción).

Esto deberá ser
 la energía potencial.

Esto es parecido al potencial
 de Newton:

$$U(r) = \frac{l^2}{2mr^2} ; \quad \left(V(r) = -\frac{\kappa}{r} \right)$$